

Maths 4

18 March 2026 20:07

✓ Pascal's Triangle

• Output:

```
char ch = 'A';
int x = ch;
System.out.println(x);
```

• Output:

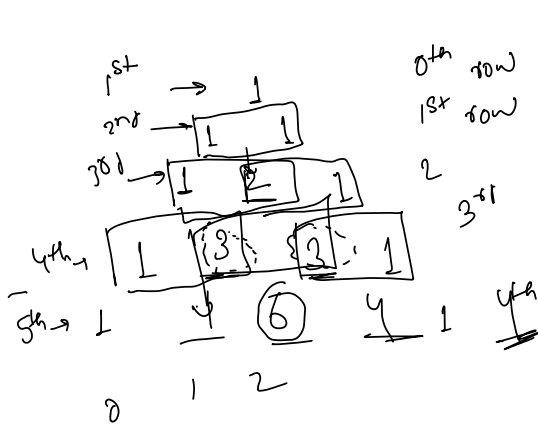
```
System.out.println('A' + 1);
```

• Output:

```
1. int a = 5;
   double b = 2;
   System.out.println(a + b);
```

```
2. System.out.println(5 + 2 + "Hello");
```

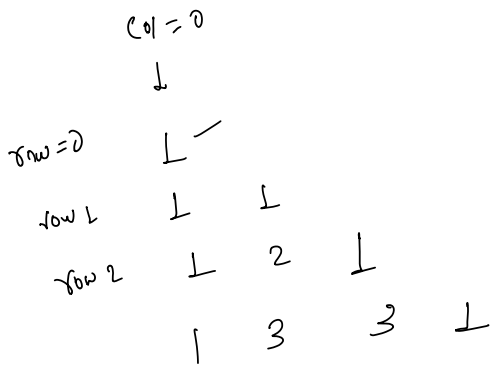
- Divide 10 by 3 and roundoff the answer upto 3 decimal Places.



Handwritten formula for combinations:

$${}^nC_2 = \frac{n!}{(n-2)! \times 2!} = \frac{4!}{(4-2)! \times 2!} = \frac{4!}{2! \times 2!} = 6$$

$${}^nC_r = \frac{n!}{(n-r)! \times r!}$$



Handwritten combination values:

$${}^3C_2 = 3$$

$${}^nC_0 = 1$$

$${}^0C_0 = 1$$

$${}^1C_0 = 1, {}^1C_1 = 1$$

$${}^2C_0 = 1, {}^2C_1 = 2$$

$${}^2C_2 = 1$$

Handwritten calculation for ${}^{10}C_3$:

$${}^{10}C_3 = \frac{10!}{7! \times 3!} = \frac{10 \times 9 \times 8 \times \cancel{7!}}{\cancel{7!} \times 3!}$$

$$= \frac{10 \times 9 \times 8}{1 \times 2 \times 3}$$

n=10

$$\frac{10-0}{0+1} \quad n=10 \quad r=3$$

$$= \frac{10}{1} \times \frac{9}{2} \times \frac{8}{3}$$

$$= \frac{10-i}{i+1}$$

$$\rightarrow \frac{10-0}{0+1} \times \frac{10-1}{1+1} \times \frac{10-2}{2+1} \times \dots \times 1$$

0
↓

3 1 3 3 1 n=3

```
for (int i = 0; i < n; i++) {
    int ans = 1;
```

```
    for (int j = 0; j < n; j++) {
```

```
        cout << ans;
```

```
        ans = ans * (n-i) / (j+1);
```

```
    }
```

Type Conversion

byte int long float double

↑

long b = 6L; double d = 3.3d
float a = 4.3f;

a, b, c, d, --- z.

```
for (char i = 'a'; i <= 'z'; i++) {
    cout << i;
```

```
}
```

for (int i = 0; i < n; i++)

